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AIM: To implement Hamming code (12,8) in C for single-bit error detection and correction, where the sender encodes text from input.txt into channel.txt and the receiver decodes the data and corrects any error.

Input.txt:

HELLO WORLD

Sender.c:

```
#include  <stdio.h>
#include  <stdlib.h>
#include <string.h>

int calculateParityBits(int dataBits)
{
   int r = 0;

   while ((1 << r) < (dataBits + r + 1))
      r++;

   return r;
}

void encodeHamming(unsigned char data, FILE *channel)
{
   int hamming[13] = {0};
   int parityPositions[] = {1, 2, 4, 8};

   int dataIndex = 0;

   // Store data bits in positions other than powers of 2
   for (int pos = 1; pos <= 12; pos++)
   {
      if (pos == 1 || pos == 2 || pos == 4 || pos == 8)
         continue;

      hamming[pos] = (data >> (7 - dataIndex)) & 1;
```

```
        dataIndex++;
    }

    // Calculate parity bits
    // P1 checks positions 1,3,5,7,9,11
    hamming[1] = hamming[3] ^ hamming[5] ^ hamming[7] ^
                hamming[9] ^ hamming[11];

    // P2 checks positions 2,3,6,7,10,11
    hamming[2] = hamming[3] ^ hamming[6] ^ hamming[7] ^
                hamming[10] ^ hamming[11];

    // P4 checks positions 4,5,6,7,12
    hamming[4] = hamming[5] ^ hamming[6] ^ hamming[7] ^
                hamming[12];

    // P8 checks positions 8,9,10,11,12
    hamming[8] = hamming[9] ^ hamming[10] ^ hamming[11] ^
                hamming[12];

    // Write 12-bit Hamming code to channel
    for (int i = 1; i <= 12; i++)
        fprintf(channel, "%d", hamming[i]);
}

int main()
{
    FILE *input;
    FILE *channel;

    char ch;
```

```
input = fopen("input.txt", "r");

if (input == NULL)
{
   printf("Error: Could not open input.txt\n");
   return 1;
}

channel = fopen("channel.txt", "w");

if (channel == NULL)
{
   printf("Error: Could not create channel.txt\n");
   fclose(input);
   return 1;
}

printf("========== HAMMING CODE SENDER ==========\n\n");

printf("Input text:\n");

while ((ch = fgetc(input)) != EOF)
{
    putchar(ch);
}

 rewind(input);

while ((ch = fgetc(input)) != EOF)
{
```

```
        encodeHamming((unsigned char)ch, channel);
    }

    fclose(input);
    fclose(channel);

    printf("\n\nHamming encoded data has been written to channel.txt\n");

    printf("\nChannel data:\n");

    channel = fopen("channel.txt", "r");

    while ((ch = fgetc(channel)) != EOF)
        putchar(ch);

    fclose(channel);

    printf("\n\nSender completed successfully.\n");

    return 0;
}
```

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Receiver.c:

```
#include  <stdio.h>
#include <stdlib.h>
#include <string.h>

int main()
{
   FILE *channel;
   char bits[100000];
   int length = 0;

   channel = fopen("channel.txt", "r");

   if (channel == NULL)
   {
      printf("Error: Could not open channel.txt\n");
      return 1;
   }

   // Read channel data
   char ch;

   while ((ch = fgetc(channel)) != EOF)
   {
      if (ch == '0' || ch == '1')
      {
         bits[length++] = ch;
      }
   }
```

```
bits[length] = '\0';

fclose(channel);

printf("========== HAMMING CODE RECEIVER ==========\n\n");

printf("Received channel data:\n%s\n\n", bits);

if (length % 12 != 0)
{
   printf("Error: Invalid Hamming data.\n");
   return 1;
}

printf("Checking for errors...\n\n");

printf("Corrected/Decoded  output:\n");

for (int block = 0; block < length; block += 12)
{
   int hamming[13];

   // Convert characters into integer bits
   for (int i = 1; i <= 12; i++)
   {
      hamming[i] = bits[block + i - 1] - '0';
   }

   // Calculate syndrome
   int p1 = hamming[1] ^ hamming[3] ^ hamming[5] ^
```

```
        hamming[7] ^ hamming[9] ^ hamming[11];

int p2 = hamming[2] ^ hamming[3] ^ hamming[6] ^
        hamming[7] ^ hamming[10] ^ hamming[11];

int p4 = hamming[4] ^ hamming[5] ^ hamming[6] ^
        hamming[7] ^ hamming[12];

int p8 = hamming[8] ^ hamming[9] ^ hamming[10] ^
        hamming[11] ^ hamming[12];

int errorPosition = p1 * 1 +
                p2 * 2 +
                p4 * 4 +
                p8 * 8;

if (errorPosition == 0)
{
    printf("[No error in block %d] ", (block / 12) + 1);
}
else if (errorPosition <= 12)
{
   printf("[Error in block %d at position %d - Corrected] ",
         (block / 12) + 1,
         errorPosition);

   // Correct the error
   hamming[errorPosition] =
      hamming[errorPosition] ^ 1;
}
else
```

```
        {
            printf("[Invalid error detected] ");
        }

        // Extract data bits
        int data = 0;

        int dataPositions[] = {
            3, 5, 6, 7,
            9, 10, 11, 12
        };

        for (int i = 0; i < 8; i++)
        {
            data = (data << 1) | hamming[dataPositions[i]];
        }

        printf("%c", (char)data);
    }

    printf("\n\nReceiver completed successfully.\n");

    return 0;
}
```

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For compilation and running:

gcc sender.c -o sender

./sender

```
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE> cd practical6
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical6> gcc sender.c -o sender.exe
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical6> .\sender.exe
========== HAMMING CODE SENDER ==========

Input text:
HELLO WORLD

Hamming encoded data has been written to channel.txt

Channel data:
000110011000110010000101010110001100010110001100100010001111010101000000110110110111100010001111100010110010010110001100110110010100

Sender completed successfully.
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical6> gcc receiver.c -o receiver
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical6> ./receiver
========== HAMMING CODE RECEIVER ==========

Received channel data:
000110011000110010000101010110001100010110001100100010001111010101000000110110110111100010001111100010110010010110001100110110010100

Checking for errors...

Corrected/Decoded output:
[No error in block 1] H[No error in block 2] E[No error in block 3] L[No error in block 4] L[No error in block 5] O[No error in block 6]  [No error in block 7] W[No error
in block 8] O[No error in block 9] R[No error in block 10] L[No error in block 11] D

Receiver completed successfully.
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical6>
```